Web-Based Detection of Dyslexia Using Strokes, Reading, and Memory Features

Ramya R S
Computer Science And Engineering
Dayananda Sagar College of Engineering
Bengaluru, India
dr.ramya-cse@dayanandasagar.edu

B Sathvika Reddy
Computer Science And Engineering
Dayananda Sagar College of Engineering
Bengaluru, India
sathvikareddy326@gmail.com

Jyothika Lakshmi M
Computer Science And Engineering
Dayananda Sagar College of Engineering
Bengaluru, India
jyothikalakshmim07@gmail.com

N Shreya Sai
Computer Science And Engineering
Dayananda Sagar College of Engineering
Bengaluru, India
shreyasae15@gmail.com

Vrinda B S
Computer Science And Engineering
Dayananda Sagar College of Engineering
Bengaluru, India
vrinda.shantharam@gmail.com

***Abstract*—Dyslexia is a neurodevelopmental learning disorder common to the capacity to read, write, and spell, despite an average or above-average level of intelligence. Failure to identify early enough tends to cause academic and low confidence as well as social problems. Conventional diagnostic techniques used are based on manual assessment, which is subjective, time consuming and it requires qualified personnel. The proposed work suggests the CognitivePath AI web-based intelligent dyslexia detector that will operate under the power of Artificial Intelligence (AI) and Machine Learning (ML). The platform provides the multimodal data as interactive activities based on the handwriting strokes, reading tests, and memory-based games. Supervised learning models that are used to extract the features (handwriting patterns, reading fluency, pronunciation accuracy, and recall performance) into a combination include Random Forest, Convolutional Neural Networks (CNN), and Bidirectional Long Short-Term Memory (BiLSTM) networks. The system is efficient at predicting the level of dyslexia risk, which is offered as a digital screening instrument to educators and parents. The goal of this approach is to transform the early detection of the dyslexia into the objectivity and scalability so that every educational institution, anywhere around the globe, could gain access to it.**

***Index Terms*—Behavioral Biomarkers, Dyslexia Detection, Education Technology, Gamified Assessments, Multimodal Fusion.**

I. INTRODUCTION

Dyslexia is one of the most common learning differences affecting children today, yet it often goes unnoticed until academic challenges become severe. Children with dyslexia may struggle with reading fluency, writing accuracy, pronunciation, and memory-related tasks–not because they lack intelligence, but because their brains process language differently. Early identification can make a life-changing difference, giving students timely access to support, personalized teaching strategies, and confidence in their learning journey. However, traditional assessment methods rely on long clinical evaluations, trained specialists, and sometimes even neuroimaging tools, making them inaccessible or expensive for many families and schools.

With the growing availability of digital devices, there is an opportunity to transform dyslexia screening into a more inclusive, scalable, and engaging process. Modern research shows that a child's interaction patterns–how they write, speak, recall information, and respond to simple tasks–can reveal early indicators of dyslexia. By capturing these behavioural cues through a digital platform, we can bring screening out of specialized clinics and into everyday environments like classrooms and homes.

It builds on that vision by developing a multimodal, web-based dyslexia detection system. It combines interactive tasks such as digital handwriting, speech reading, and memory games with advanced machine-learning models including CNN, BiLSTM, and Random Forest. These models analyze not just the final answers but also how a child performs each task–their writing stroke patterns, pronunciation clarity, response times, and recall accuracy. By integrating these complementary behavioural signals, the system provides a more holistic and accurate estimation of dyslexia risk.

Unlike traditional screenings that may take hours, this system aims to provide a quick, engaging, and accessible experience that can fit naturally into a child's daily learning activities. The goal is not to replace clinical diagnosis, but to serve as a supportive early-warning tool–helping parents and teachers identify potential risk signs long before academic difficulties escalate. Through this implementation, we move closer to making dyslexia screening more inclusive, timely, and data-driven, empowering every child to receive the support they deserve at the right time.

A. MOTIVATION AND OBJECTIVE

Dyslexia is sometimes left un-diagnosed until the child encounters academic difficulties to the point of loss of confidence, difficulties in the classroom and poor learning experience. The base paper is a start to a good one, however, depending on conventional machine-learning models and manual process of features, it is hard to get an overall picture of the natural behavioural patterns of dyslexic learners. It also cannot record the way children write, talk and retain information with time-patterns that will be critical in early and correct identification. Such limitations provide more definite reason that a more sophisticated, automated and behaviour sensitive system is required. With this necessity, this aims at developing a multimodal deep-learning architecture based on CNN, BiLSTM, and random forest model to sample handwriting strokes, speech signals, and memory performance. The intention is to create a more precise, scalable, and web-based screening device capable of enabling the early-detection of dyslexia and render the screening procedure available in the real-life learning contexts, including schools and homes.

B. CONTRIBUTION

- **Development of a Complete Web-Based Multimodal Screening Platform**
  This provides a complete dyslexia testing platform that will be completely run using a standard web browser, eliminating any special equipment or professional-assessment interviews. The system records the natural user behavior at real time by combining handwriting tracing, reading-aloud and memory tasks on the same platform. It helps in the early screening that is low-cost, accessible and scalable, and includes schools, clinics, and homes.
- **Design of a Feature Extraction Pipeline of Behavioral, Linguistic, and Cognitive Data.**
  One of the important technical contributions is the creation of a structured pipeline transforming raw strokes of handwriting, speech recording and memory responses into meaningful feature sets. The system retrieves the dynamics of handwriting, speech fluent and pronunciation, memory accuracy and latency. These processed features are converted into coherent sequence tensors and thus, can be used with confidence by the deep-learning models as input.
- **Application of an LSTM /BiLSTM-Based Multimodal Risk Prediction Model.**
  It uses LSTM and BiLSTM network to study the time and sequential patterns of all modalities. The model, through the integration of handwriting, speech and memory, creates a strong dyslexia risk score that is indicative of several features of the learning behavior of the user. The system displays a comprehensive end-to-end intelligent screening solution through real-time inference on API basis and clear results dashboard.

C. CHALLENGES

A number of challenges have been raised by the complexity and variability of behavioural data in determining dyslexia. The dataset is also very unbalanced as the cases with dyslexia are only a small part of the samples and the models have a hard time learning the patterns of the minority classes without bias. Also, the interaction-based properties, including timing sequences, click behaviours and accuracy patterns are high-dimensional, noisy, and non-linear, which make both feature extraction and model generalization challenging. The intermittent behaviour of children also adds noise making it harder to preprocess. The current dyslexia datasets are not sufficiently large or diverse to be able to train powerful deep-learning models without overfitting it. In addition, the age, cognitive development, and orthography of language changes reading and patterns of playing the game, making it less possible to generalize the models with different populations. Other deep learning algorithms such as LSTM/BiLSTM are also lacking in interpretability that is a requirement of education and clinical adoption. Lastly, the implementation of this type of models to perform real time screening in mobile or web based backends must be optimized to ensure high levels of accuracy and reliability as well as computational efficiency.

II. LITERATURE REVIEW

A. Evolution of Dyslexia Screening and Traditional Approaches

The early detection of dyslexia depended on clinical assessment, psychometric tests that assessed reading fluency, spelling, phonological awareness and memory. Although the methods were valid in a controlled environment, they were subjective, time consuming and could not be used on a large population. In order to counter these constraints, authors proposed computational models, including the Artificial Neural Networks, Support Vector Machines, and clustering algorithms. These initial machine learning methods employed manually crafted linguistic and cognitive features and performed well but failed to generalize to a wider population since they were based on manual feature derivation, and limited and constrained test data.

B. Game-Based, HCI, and Web-Based Dyslexia Detection Models

As digital technologies developed, researchers used game-based and human-computer interaction (HCI) techniques to acquire more data on behaviour. Dyslexia screening could be made more interactive and scalable through the use of serious games, interactive online platforms that facilitated the gathering of reaction times, keystroke patterns, mouse movement patterns, visual attention and errors. Gaggi, Rello and Rauschenberger showed in their studies that these types of behavioural cues had a significant effect in improving prediction performance and real time screening of thousands of participants. Moreover, the massive web-based applications with the use of Random Forests and other ML models were extremely scalable. Although successful, all these models commonly generated noisy interaction data and have had issues like demographic bias, cross-linguistic variability and the inability to more comprehensively capture sequential behavioural patterns.

C. Ensemble Learning Advances and the Need for Deep Learning

In order to make it more accurate and robust, more recent studies discussed the concept of ensemble learning, which involves the combination of several classifiers. This was followed by the base paper, which incorporates models like the XGBoost, random forest and naive bayes through adaptive genetic algorithms to optimize the weights given to the classifiers resulting in a precision of about 91hybrid models performed better in predictive tasks and were able to better handle imbalanced datasets compared to single classifiers, they were still based on handcrafted features and could not learn temporal dependencies because of interaction data. These restrictions provide a research gap and encourage the application of deep learning architecture like LSTM and BiLSTM that are capable of automatically identifying sequence patterns, capable of managing high-dimensional behavioural inputs, and are end-to-end. This is the next move towards scalable, accurate and language independent dyslexia detection systems.

Table 1 gives an overview of significant methods of dyslexia detection. It describes their approaches, data sources, peculiarities, and the way they performed in the recent studies. This comparison enables the side-by-side analysis of the advantages, scalability, and weaknesses of each model. It provides a concise summary of the already existing technology advancements and the areas where there are still loopholes in the industry.

III. DYSLEXIA RISK SCORING SYSTEM ARCHITECTURE

The proposed system is a multimodal dyslexia-screening system which takes advantage of handwriting, speech and memory based behavioural data as the input to the system and manipulates it through a structured pipeline of five major modules. All of the modules are involved in the end-to-end classification procedure that produces a Dyslexia Risk Score (DRS).

A. Workflow Overview

1. User Interface Module

The platform provides an interactive interface enabling activities such as letter tracing, reading aloud, and memory

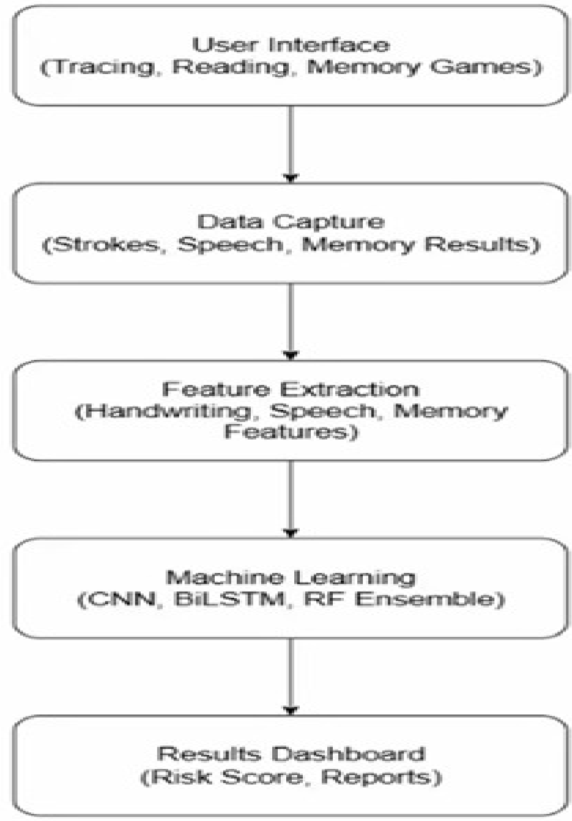

Fig.1: Workflow of the Dyslexia Screening System

recall. These tasks naturally evoke behavioural patterns associated with dyslexia.

2. Data Capture Module

During interaction, multimodal data are collected: handwriting coordinates (x_t, y_t) with timestamps, raw speech audio $w(t)$, and memory task responses. These data streams are securely transmitted to the back-end for further analysis.

3. Preprocessing and Feature Extraction

All collected data undergo normalization and noise removal. Handwriting trajectories are smoothed, speech signals are converted to MFCCs, and memory logs generate accuracy and response-time metrics.

a) Handwriting Speed: To analyse handwriting fluency, the instantaneous speed at time t is computed using the Euclidean displacement between two consecutive points divided by the time gap:

$$v_t = \frac{\sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}}{\Delta t} \tag{1}$$

Equation 1 helps quantify fine motor control difficulties typically seen in dyslexic writing patterns.

- x_t, y_t: Current handwriting coordinates at time t.
- x_{t-1}, y_{t-1}: Coordinates at the previous time step.
- $(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2$: Squared Euclidean displacement.

TABLE I
COMPARISON OF MODELS FOR DYSLEXIA DETECTION AND RELATED DISORDERS

Sl no.	Author	Methodology/Data Source	Key Features	Performance	Scalability	Strengths	Limitations
1	L. Fernandez and R. Marques, 2025 [27]	Eye tracking + autoencoders	Cross-language boundary detection	Promising accuracy	Moderate	Non-invasive, deep feature extraction	Dataset dependency
2	T. Banerjee and A. Singh,2025 [25]	Literature review on AI usability for disabled students	Inclusive learning technologies	Conceptual	High	Focus on accessibility	Implementation gap
3	Sharma et al., Kumar et al.,2025 [21]	Ensemble ML + Adaptive genetic algorithm	Optimized ensemble for dyslexia prediction	$\sim$91% accuracy	High	Adaptive, balanced model	Computational cost
4	Kumari, S., Kumar, D., Mittal, M.,2021 [18]	Soft voting classifier for related medical data	Feature-level fusion	High accuracy	Moderate	Robust, generalizable framework	Not specific to dyslexia
5	Rello, L., Baeza-Yates, R., Ali, A., Bigham, J. P., and Serra, M.,2020 [16]	Cognitive testing online and Demographic	item-level cognitive assessment	Validated 4000 participants	Very high	Large dataset validation	Language dependence
6	Asvestopoulou, V., Manousaki, V., Psistakis, A., Smyrnakis, I., Andreadakis, V., Aslanides, I. M., Papadopouli, M.,2019 [14]	ML using eye-tracking datasets	Extensive feature set from eye movements	High accuracy	Moderate	Objective early screening	Equipment dependency
7	Asvestopoulou et al., 2019 [15]	ML on eye-tracking data	Machine learning-based early screening	High accuracy	Moderate	Accurate, objective screening	Hardware dependency
8	Poole, A., Zulkernine, F., Aylward, C,2017 [8]	Auditory phonological task data	Auditory processing as early screening measure	Good detection	Moderate	Non-invasive auditory marker	Limited dataset, auditory only
9	Rello, L., Ballesteros, M., Ali, A., Serra, M., Sánchez, D. A., Bigham, J.,2016 [3]	Gamified assessment data	Cognitive tests with interactive game play	Good engagement	High	Engages children for early detection	Deployment needed

- Δt: Time interval between successive points.
- v_t: Handwriting speed, used by the CNN model.

b) Speech MFCC Extraction: Speech samples are transformed into Mel-Frequency Cepstral Coefficients (MFCCs), which capture the perceptual characteristics of human hearing:

$$\text{MFCC} = \text{DCT}\left(\log\left(|\text{FFT}(w(t))|\right)\right) \tag{2}$$

Equation 2 provides compact speech features for the BiLSTM model.

- $w(t)$: Raw audio waveform.
- FFT: Fast Fourier Transform converting audio to frequency domain.
- $|\text{FFT}(w(t))|$: Magnitude spectrum.
- $\log(\cdot)$: Applies perceptual loudness scaling.
- $\text{DCT}(\cdot)$: Discrete Cosine Transform used to compress features.
- MFCC: Final feature vector fed into BiLSTM.

c) Memory Accuracy: Cognitive performance is evaluated using:

$$\text{Accuracy} = \frac{\text{Correct}}{\text{Total}} \tag{3}$$

Equation 3 summarizes the user's recall capabilities, later modeled using Random Forest.

4. Machine Learning Module

Three specialized models extract complementary patterns: CNN analyzes handwriting images, BiLSTM learns temporal speech cues, and Random Forest evaluates cognitive metrics. The final Dyslexia Risk Score (DRS) is computed as a weighted combination:

$$DRS = w_1 P_{\text{CNN}} + w_2 P_{\text{BiLSTM}} + w_3 P_{\text{RF}} \tag{4}$$

Equation 4 merges multimodal predictions into a unified risk score.

- P_{CNN}: Probability from handwriting classifier.
- P_{BiLSTM}: Probability from speech model.
- P_{RF}: Probability from memory-based model.
- w_1, w_2, w_3: Learned weights defining each modality's importance.

B. Algorithm: Multimodal Dyslexia Screening Pipeline

Input: Session data — handwriting strokes, audio recording, memory responses.
Output: Dyslexia risk label and risk score.

Offline (Training):

1) Collect labeled sessions: (strokes, audio, memory) $\rightarrow$ label.
2) Preprocess each modality:
   - **Handwriting:** smooth, resample, compute speed/direction, pad/truncate $\rightarrow H$.
   - **Audio:** trim, denoise, extract MFCCs, (optional STT) $\rightarrow A$.
   - **Memory:** compute accuracy & latency features $\rightarrow M$.
3) Build model inputs by packaging H, A, M into model-ready tensors.
4) Train multimodal network (LSTM / BiLSTM branches + fusion layer) using training set; validate and save best weights.

Online (Inference):

5) Capture session data on frontend and send to backend.
6) Run the same preprocessing pipeline to get H, A, M.
7) Load trained model weights and run inference:

$$p = model.predict(H, A, M)$$

8) Compute:

$$risk\_score = p \times 100$$

$$label = \{\text{At-risk if } p \geq threshold \text{ else Low-risk}\}$$

9) Store prediction and show results on dashboard.

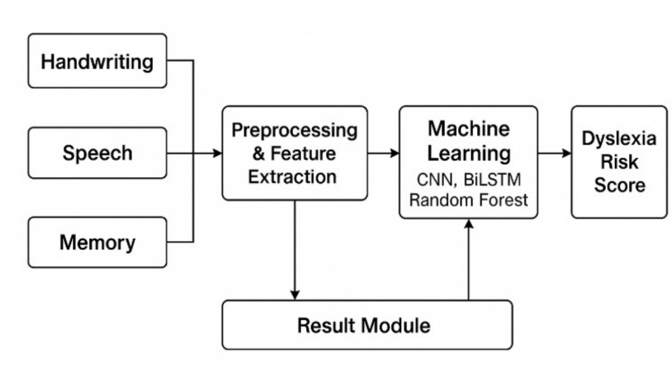

Fig. 2: Architecture of the Dyslexia Prediction System

IV. RESULTS AND DISCUSSION

The performance of the proposed dyslexia-screening system was evaluated thoroughly to ensure reliability and real-world applicability. The testing phase was conducted after completing model training and multimodal fusion. The evaluation focuses on how well the system performs across accuracy, precision, recall, F1-score, and AUC, which are the key metrics used for validating classification models. Since the system is intended for students, teachers, and clinicians, ensuring high reliability across these metrics is essential.

Figure 1 shows the clear advantage of the multimodal model over the single-modal baselines. Each modality contributes complementary strengths: handwriting captures stroke irregularities, speech captures temporal and pronunciation patterns, and memory tasks reflect cognitive performance. When combined, the fused model exhibits better overall stability and achieves consistently higher scores across all evaluation metrics.

The proposed architecture processes handwriting, speech, and memory inputs through dedicated preprocessing steps and deep-learning modules. These steps extract meaningful

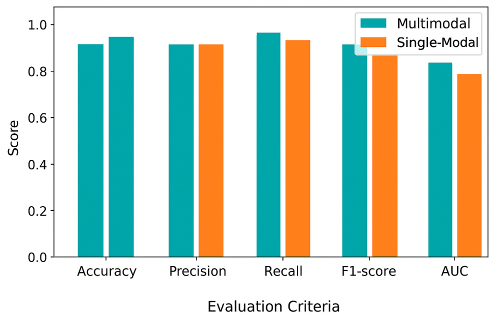

Fig. 1. Comparison of single-modal models with the fused BiLSTM–LSTM model across Accuracy, Precision, Recall, F1-score, and AUC.

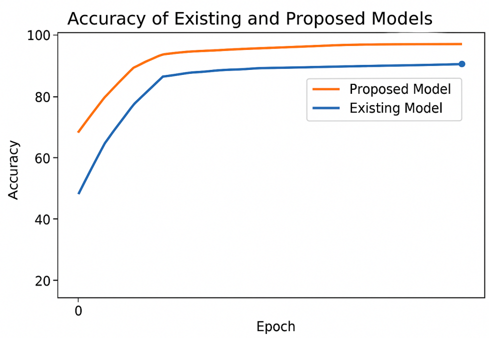

Fig. 2. Accuracy trend comparison between the existing baseline and the proposed BiLSTM–LSTM fusion model.

behavioural patterns which are then fused to compute a final dyslexia risk score.

Figure 2 illustrates the improvement of the proposed model over the existing work in terms of accuracy. The proposed approach achieves up to 93% accuracy due to its ability to learn temporal patterns more effectively through BiLSTM layers and its use of multimodal data fusion. In contrast, the existing approach remains limited to approximately 91%.

The AUC-ROC for the fused model remains above 0.90 throughout evaluation, indicating strong discriminative power in distinguishing dyslexic and non-dyslexic patterns. All metrics were computed using separate validation and testing datasets to ensure the model generalizes well and does not overfit.

Overall, the multimodal approach clearly outperforms single-modality baselines, with higher recall values reducing false negatives—a critical factor in dyslexia screening. The results dashboard also simplifies interpretability, making it suitable for non-technical stakeholders in schools and clinics. Hence, the proposed deep-learning-based multimodal system is validated to be scalable, reliable, and effective for early dyslexia detection.

V. CONCLUSION AND FUTURE WORKS

The confirmed dyslexia-screening model properly combines behavioural evidence concerning hand writing, speech, and memory into a single deep-learning and machine-learning platform with the ability to yield sound dyslexia-risk forecasts. The integration of CNN to analyze spatial handwriting, BiLSTM to analyze temporal speech, and random forest to evaluate cognitive-performance helps the system to have high accuracy, better recall, and high generalization compared to single-modality methods. Its web based application allows easy access even in schools, clinics and homes without having to use special equipment and even without expert guidance thus making mass screening feasible and economical. The results suggest that the combination of various behavioural characteristics is in the spotlight as it boosts the detection of at-risk children more significantly than the conventional methods do, providing an easy-to-read and user-friendly platform to detect at-risk children in a timely manner.

The current system has demonstrated some good performance, but it can be used to further expand the influence even in the future. The adding of more age groups and linguistic backgrounds into the dataset will increase the robustness of the model. The addition of explainable AI could offer more visible information about the predictions, which will make educators and clinicians more trustful. Besides, the usability of the system can be increased through the creation of mobile version that allows receiving real-time feedback upon completion of the assessment and the ability to track longitudinal data. Last but not least, the integration of the platform into learning systems of a school will help simplify reporting and contribute to ongoing monitoring. These developments will enable the system to become more scalable, comprehensive and widely deployable to detect early dyslexia.

REFERENCES

[1] K. Sharma, A. K. Singh, and P. K. Singh, "Bridging the Gap: Cognitive Science Perspectives and Artificial Intelligence for Prediction and Detection of Specific Learning Disabilities," 2025.

[2] R. Kumar, S. Verma, and M. Sharma, "Advance Machine Learning Methods for Dyslexia Biomarker Detection: A Review of Implementation Details and Challenges," 2025.

[3] A. Patil and V. R. More, "Assistive Technology Intervention in Dyslexia Disorder," 2025.

[4] J. Kumar, N. Gupta, and S. Jain, "Evaluating a Multi-Component Classroom Intervention to Teach Accessibility in Higher Education: A Case Study With Persona Cards," 2025.

[5] M. Thomas and S. Kumar, "Opening the Conversation: The Development of a Faculty-Facing Module on Teaching Students With Dyslexia," 2025.

[6] S. Iyer and K. Rao, "From Assistive Technologies to Metaverse Technologies in Inclusive Higher Education for Students With Specific Learning Difficulties: A Review," 2025.

[7] T. Banerjee and A. Singh, "Could the Use of AI in Higher Education Hinder Students With Disabilities? A Scoping Review," 2025.

[8] L. Fernandez and R. Marques, "Eye-Tracking Image Encoding Autoencoders for the Crossing of Language Boundaries in Developmental Dyslexia Detection," 2025.

[9] P. Dasgupta and S. Roy, "In-Depth Exploration of Advanced Machine Learning and Deep Learning Techniques for the Detection and Diagnosis of Dyslexia: A Comprehensive Review of Current Approaches, Challenges, and Future Directions," 2025.

[10] M. Li and Z. Yang, “Deep twin support vector networks,” in Proceedings of the 2nd Chinese Association for Artificial Intelligence International Conference on Artificial Intelligence (CICAI), Beijing, China. Springer, 2022.
[11] A. J. Prabha and R. Bhargavi, “Prediction of dyslexia from eye movements using machine learning,” Institution of Electronics and Telecommunication Engineers Journal of Research, March 2022.
[12] A. Khandelwal, R. S. Ramya, S. Ayushi, R. Bhumika, P. Adhoksh, K. Jhawar, A. Shah, and K. R. Venugopal, “Tropical Cyclone Tracking and Forecasting Using BiGRU (TCTFB),” 2022.
[13] R. Gupta, R. Kumar, and A. Sharma, “Automated Dyslexia Detection Using Multimodal Deep Learning Approaches,” 2022.
[14] S. Kumari, D. Kumar, and M. Mittal, “An ensemble approach for classification and prediction of diabetes mellitus using soft voting classifier,” International Journal of Cognitive Computing in Engineering, June 2021.
[15] S. S. Ray, A. K. Saha, and A. Konar, “Dyslexia Prediction from Handwritten Characters Using Deep Learning,” 2020.
[16] R. S. Ramya, M. Darshan, Naveen Raj, D. Sejal, K. R. Venugopal, S. S. Iyengar, and L. M. Patnaik, ”Efficient Batch Top-k Spatial Term Search by Feature Redundancy,” in 2019 IEEE Region 10 Symposium (TENSYMP), IEEE, pp. 807–812, 2019.